



Mapping Urban Development & Gentrification In Dallas, TX

EPPS 6302: Methods of Data Collection and Production

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The Displacement Crisis

- **Accelerated Gentrification Driven by Economic Investment:** Dallas is a prime target for urban redevelopment and real estate speculation due to emerging economic investment, accelerating gentrification.
 - Gentrification is defined as the transformation of lower-income urban areas via an influx of wealthier residents and capital through targeted investments.
- **Dramatic Shift in Housing Affordability:** The crisis is widespread, with one in five Dallas neighborhoods in dynamic or late stages of gentrification.
 - This economic shift is reflected by average home sale prices in these areas surging from approximately \$55,000 (2011–2012) to \$238,000 (2021–2022).
- **Disproportionate Impact on Vulnerable Communities:** These changes disproportionately impact Latino and Black communities in historically marginalized neighborhoods, increasing their risk of displacement and cultural erasure.
 - We specifically investigate the correlation between gentrification, urban investment, and resulting demographic shifts in these areas.



Research Question

“Is there a relationship between the increase of the median housing costs within the City of Dallas and the change in the racial and ethnic demographics over the last 30 years? “

Hypothesis

H_0 : There is no linear relationship between percentage change in median household income and the change in owner-occupied housing share across Dallas census tracts.

H_a : There is a linear relationship between percentage change in median household income and the change in owner-occupied housing share.

Theoretical Framework

01

Data Produced

Following Salganik (2019), we treat administrative records (e.g., Census, Zillow) not as “raw truths” but as “ready-made” data used for bureaucratic purposes, which require careful curation.

02

Algorithmic Culture

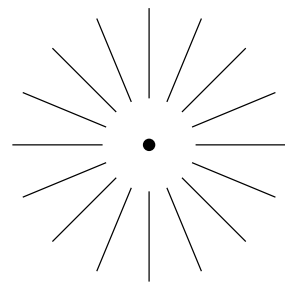
Adopting Breiman’s (2001) “Algorithmic Modeling Culture” we use predictive variables and complex inputs to understand the “black box” of urban change.

03

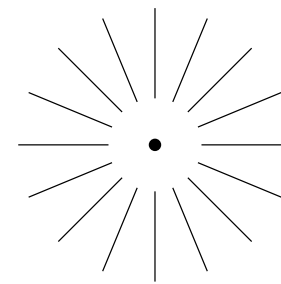
Hybrid Strategy

We blended the large-scale API (Macro) data with granular fitting (Micro) to create a custom research dataset specific to the City of Dallas boundaries.

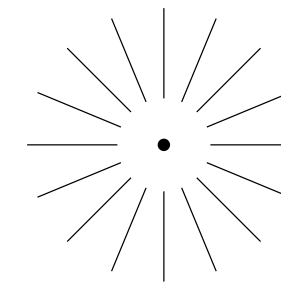
Reproducible Pipeline



R Extraction: tidycensus and fredr packages pull 30 years of raw data

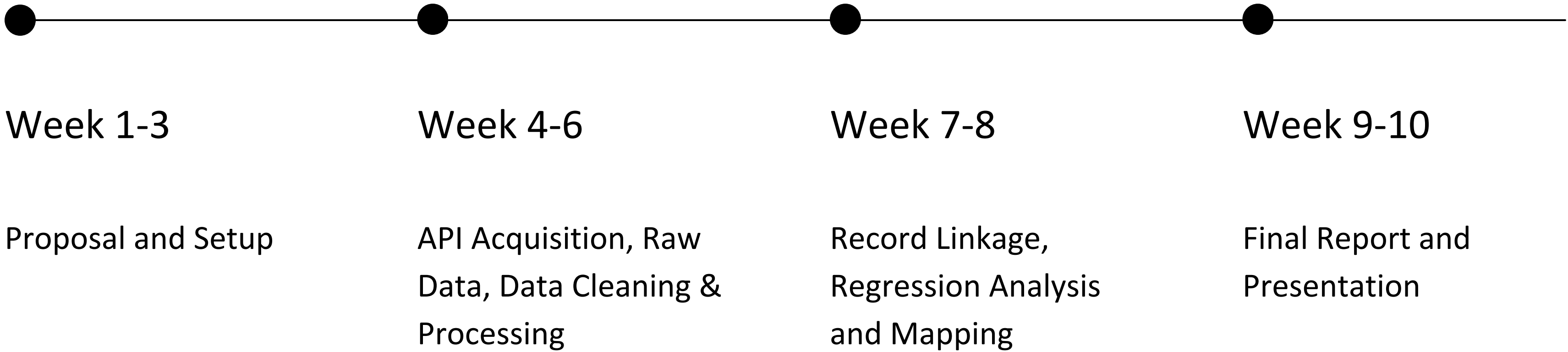


Cleaning: Scripts filtered the broad MSA data down to Dallas tracts.



Production: Record linkage harmonizes decennial snapshots with continuous housing indices.

Timeline



Data Production Strategy

01

Demographic Data

Source: Census & ACS (1990-2022)

We standardized variable names across 3 decades to track shifting racial compositions. Raw counts were converted to percentage shares to allow for direct comparison across years with varying total populations)

02

Housing Data

Source: FRED and Zillow Indices

We aggregated monthly ZHVI indices into annual averages to match census years. Inflation adjustment were applied to 1990 and 2000 figures to (to 2025 dollars) to ensure the 30-year price comparison was valid

Analysis Process

Step 1- Data Scraping

Step 2- Data Cleansing

Step 3- Merge All Datasets Into One Panel

Step 4- Create Change Metrics (2010 → 2020)

Step 5- Statistical Modeling

Step 6- Correlation Analysis

Step 7- Prepare Spatial Data

Step 8- Visualization

Variables

Dependent variable (Y)- change in owner-occupied housing share

independent variable (X)- a percentage change in median household income

To test the relationship between the change variables over the 2010-2020 period

Descriptive Variables

1. Decennial Census (2000, 2010, 2020)

- Geographical level: Census Tract
- Variables used:
 - geoid
 - name
 - year
 - total_pop
 - housing_units

3. Zillow ZHVI (City-Level)

- Geographical level: City of Dallas
- Variables used:
 - year
 - zhvi_mean_city

2. ACS 5-Year Estimates (2010–2020)

- Geographical level: Census Tract
- Variables used:
 - geoid
 - year
 - median_hh_income (renamed median_income)
 - owner_share
 - renter_share

Finalize unified dataset

- geoid
- name
- year
- total_pop
- housing_units
- median_income
- owner_share
- renter_share
- zhvi_mean_city

Why We Only Use the 2020 Shapefile for 2010–2020 Change Maps

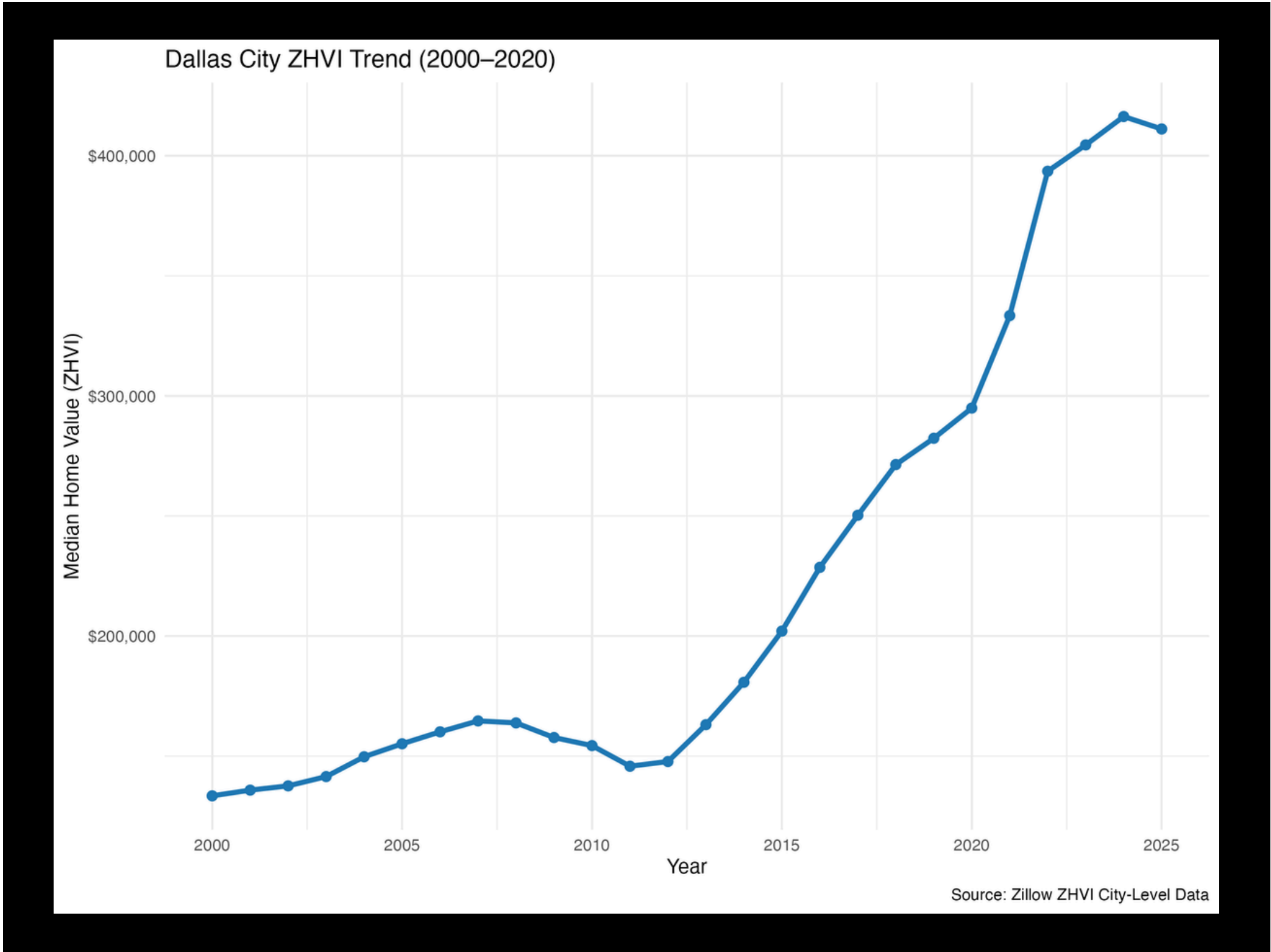
- Census tracts change every decade: they can split, merge, or shift.
- 2000 tracts $\neq$ 2010 tracts $\neq$ 2020 tracts.
- Their GEOIDs and shapes are not the same.
- Change calculations require both years to be mapped onto one reference geography.
- Modern research (Urban Institute, LTDB, NCDB, city planning) standardizes everything to 2020 tracts.
- Our analysis already uses 2020 tracts as the unit of analysis.

Key Findings

From calculated the change metrics for 628 Dallas Census Tracts that had complete data for the 2010 and 2020 periods, the results are as follows;

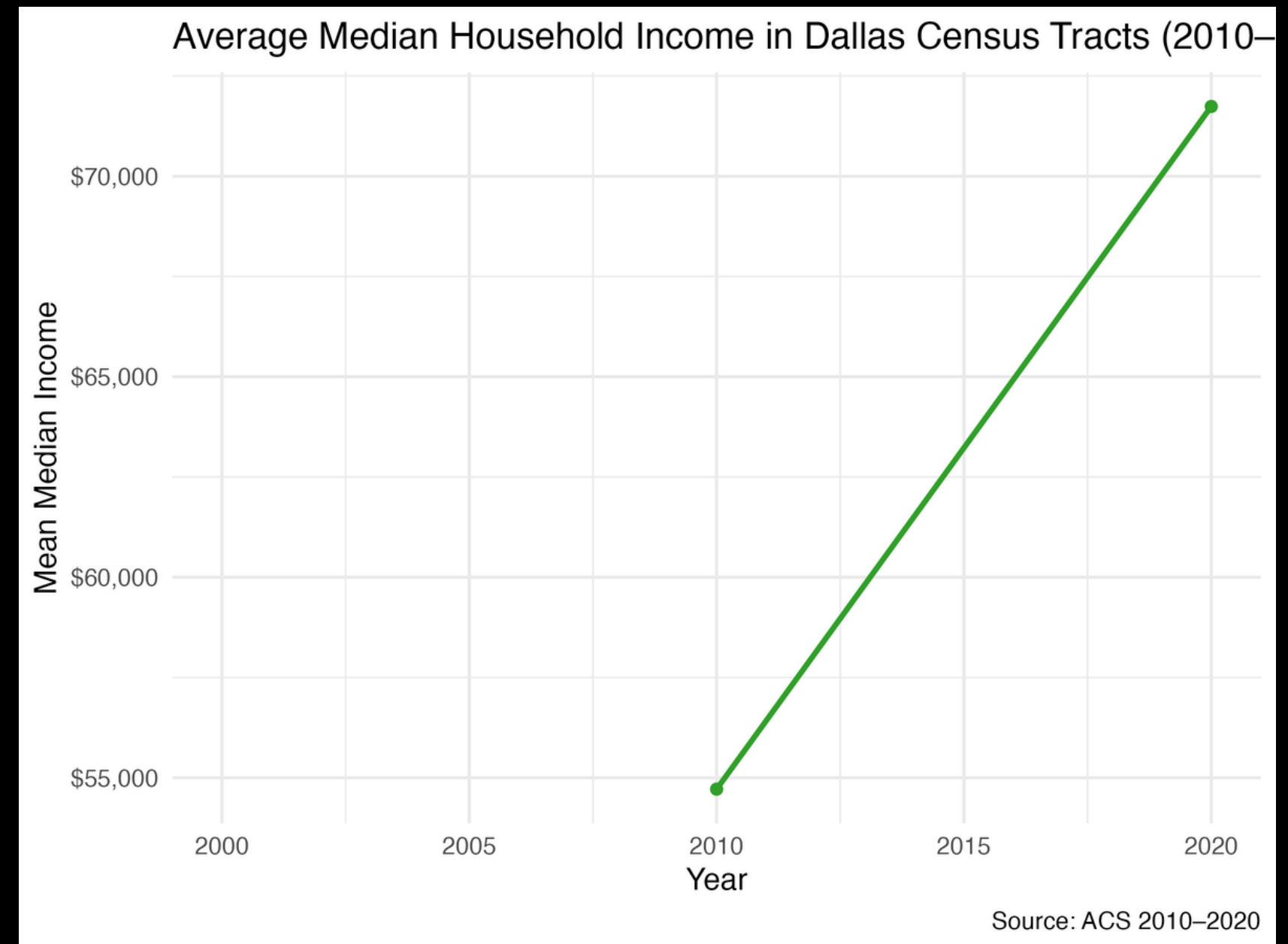
Statistic	Δ Owner Share (pp)	% Δ Median Income (%)
N	628	628
Mean	0.88 pp	32.12%
Median	1.13 pp	29.80%
Standard Deviation	9.99 pp	23.36%
Minimum	-40.69 pp	-16.09%
Maximum	44.57 pp	200.08%

City-Level ZHVI Trend shows the general upward trajectory of housing values in Dallas between 2000 and 2025, which confirms that housing cost pressure was a macro factor across the study period.

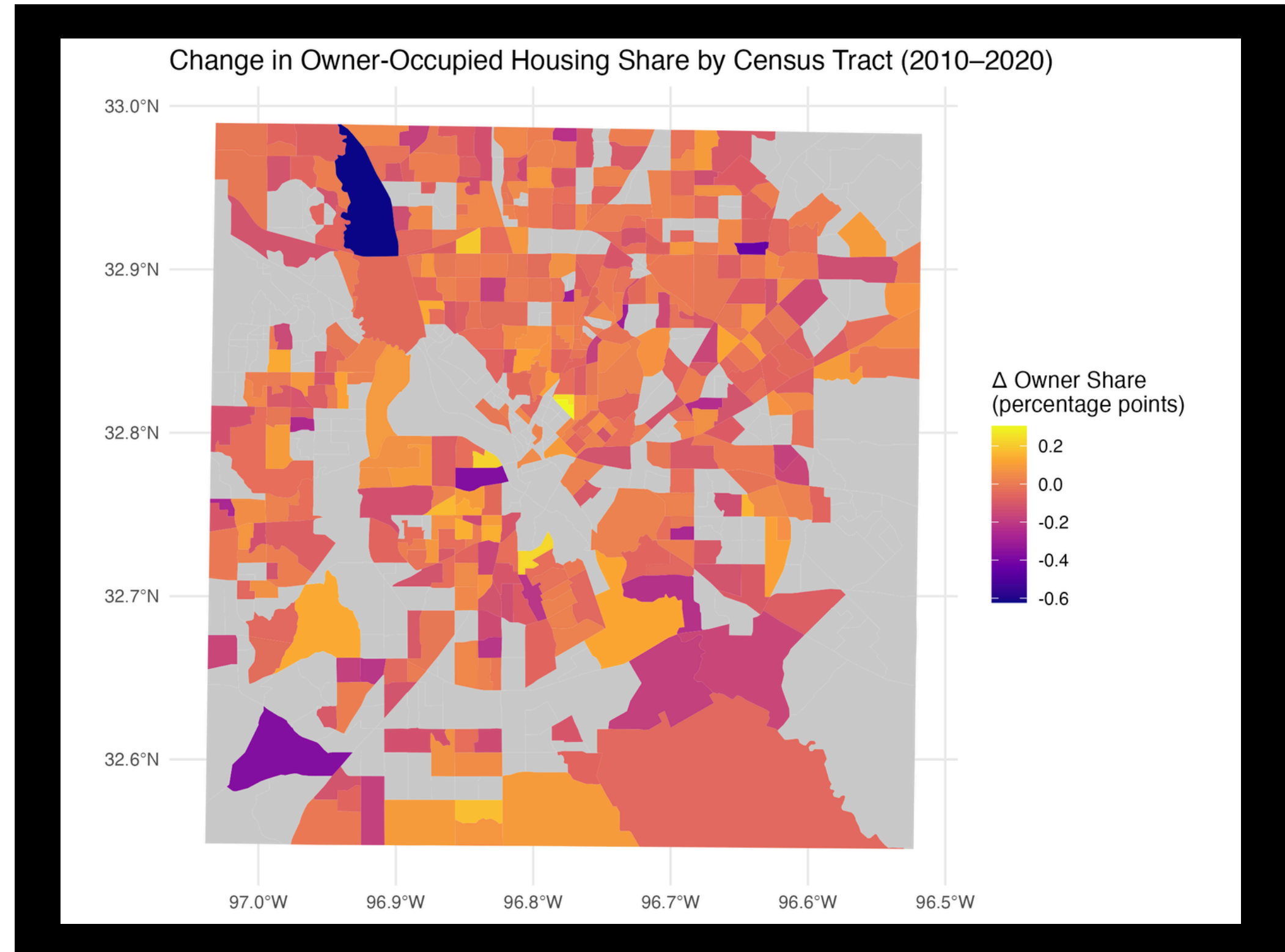


Average Tract Income

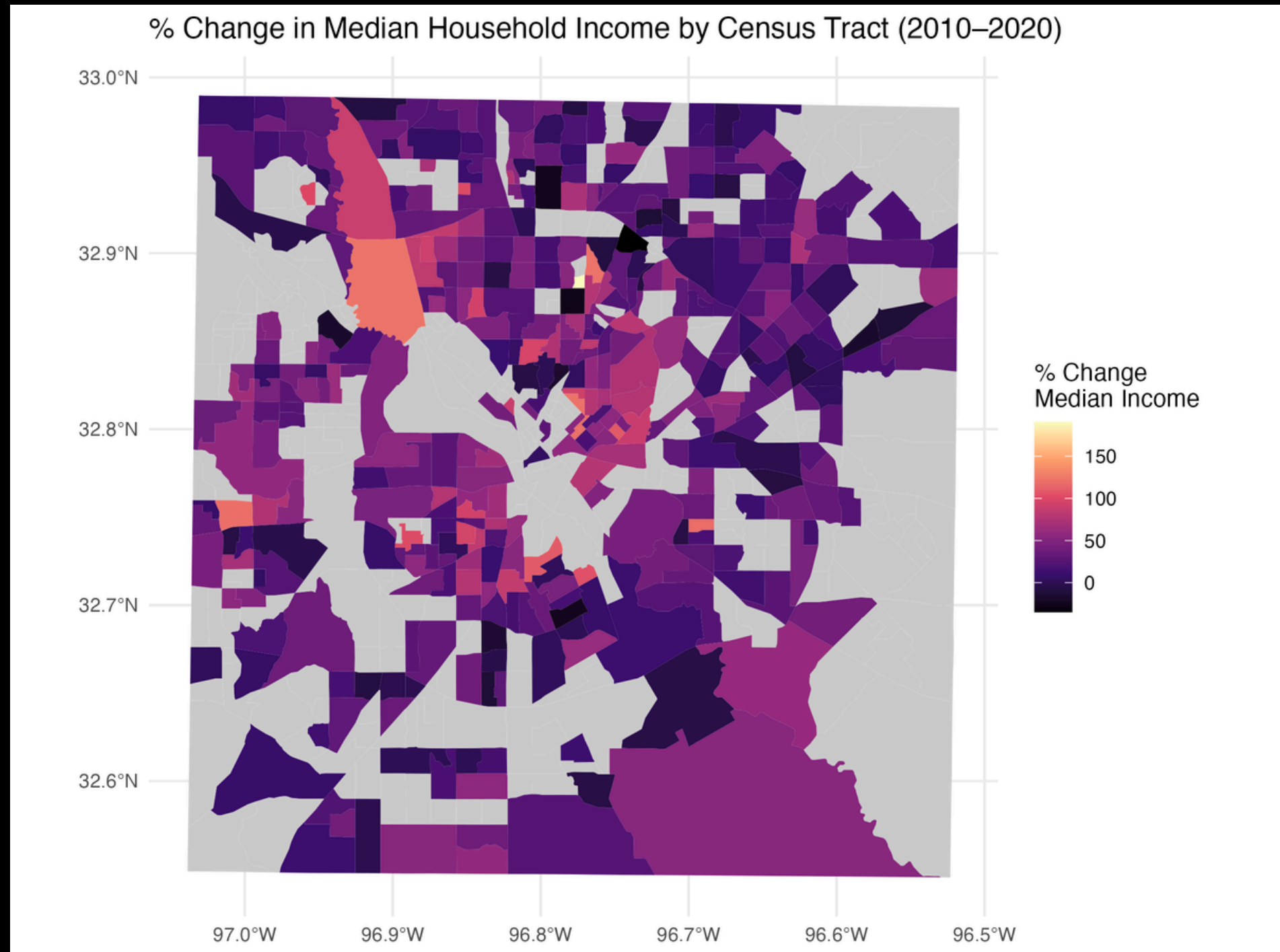
Trend shows the average median household income across the studied tracts increasing substantially from 2010 to 2020, rising from approximately \$55,000 to over \$75,000 (based on the summary data).



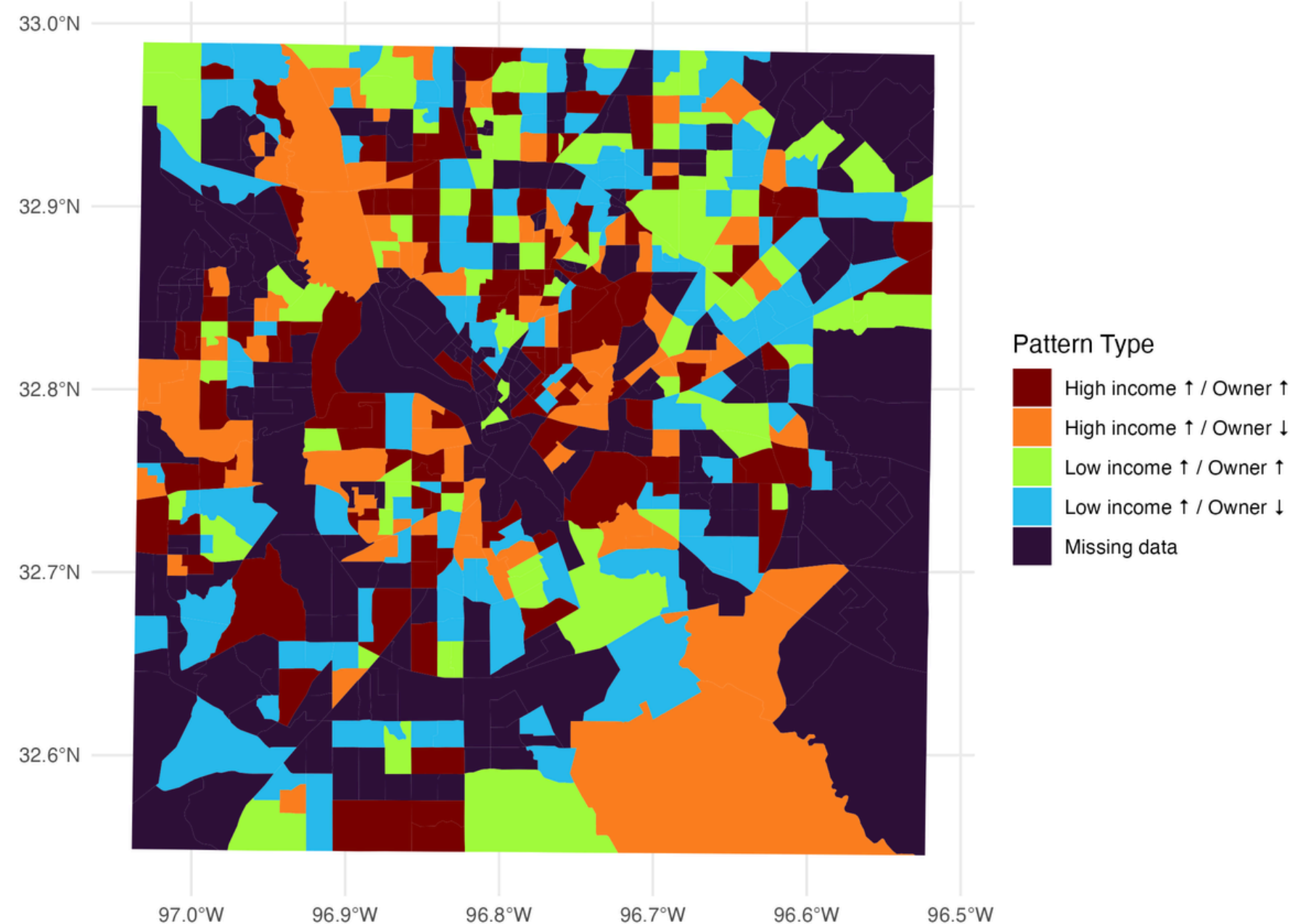
The Change in Owner-Occupied Housing Share above shows that lost owner share and areas that gained it. Losses are concentrated in specific, often low-income, inner-city neighborhoods, while gains are clustered in the suburbs and in some redeveloped urban corridors.



The Percentage Change in Median Household Income map visually displays the areas that experienced the largest economic shifts. The tracts with the highest percentage increases in median income are often those that had lower initial incomes, consistent with patterns of economic gentrification.



Income & Owner-Occupied Change Pattern by Census Tract (2010–2020)



- Gentrification Pattern Types are determined using a median-split approach.
- This method categorizes tracts based on whether their change in income and owner share was above or below the county median.

1. High income ↑ / Owner ↑

- a. Represents areas of significant affluence and stabilization.
- b. Indicates growth in both median income and owner-occupied housing.

2. High income ↑ / Owner ↓

- a. Consistent with “renter-gentrification” or displacement vulnerability.
- b. Characterized by rapidly increasing income, coupled with a loss of owner-occupied housing.
- c. These tracts represent the largest demographic shifts

Regression Analysis (OLS) Results

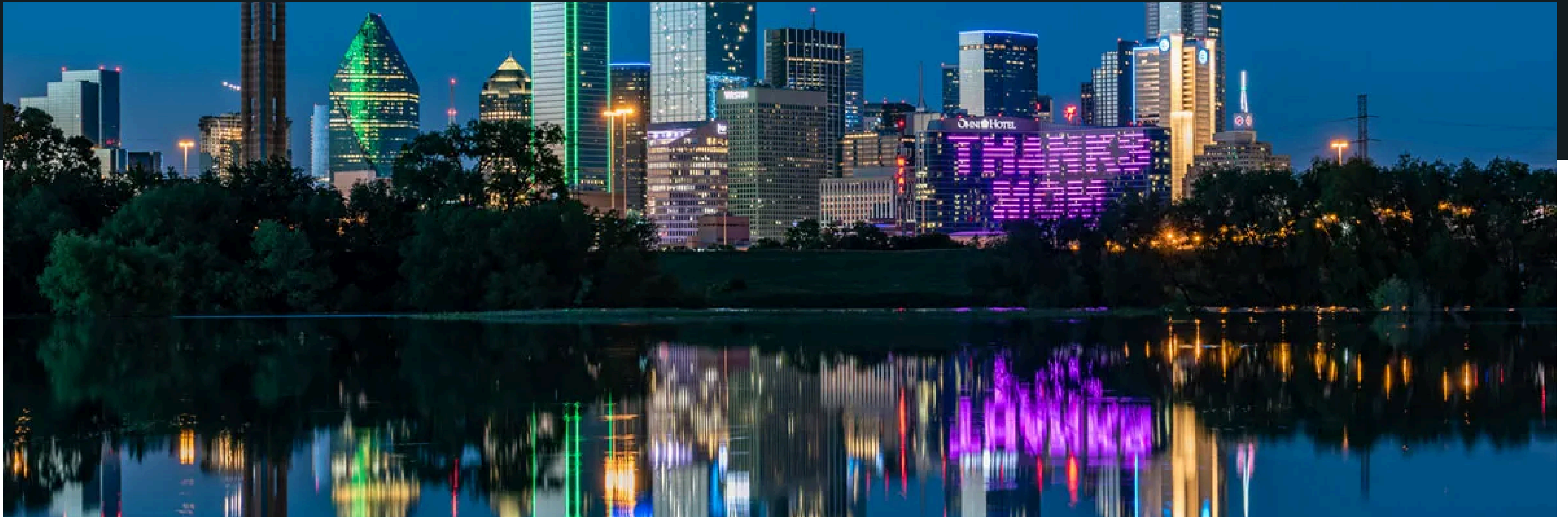
To test the relationship between the change variables over the 2010-2020 period

The result is that we cannot reject the null hypothesis so Changes in income do not predict changes in owner-occupied housing share.

Term	Estimate (β)	Std. Error	t-statistic	p-value
(Intercept)	1.157	0.828	1.397	0.163
pct_change_median_income	-0.008	0.024	-0.347	0.729

Policy Implications & Discussions

The Future of Dallas



Findings

- **Analysis Focus:** The research centered on the relationship between economic growth and housing structure shifts across Dallas census tracts between 2010 and 2020.
- **Uneven Change Observed:** Visualization of trends confirmed that Dallas experienced a highly uneven change, emphasizing the heterogeneity in impacts across neighborhoods.
 - Areas of rapid affluence were concentrated.
 - These economic hotspots were often associated with tracts experiencing rapid shifts in housing tenure.
- The central hypothesis proposing a statistically significant linear relationship between the percentage change in median income and the change in owner-occupied share was refuted by the Ordinary Least Squares (OLS) regression model.
- **Income as a Predictor:** The degree of change in the tract's median income was a negligible predictor of whether the owner-occupied share increased or decreased during the 2010–2020 decade

Implications

- This finding indicates that simple income growth is not the sole, dominant driver of shifts in owner-occupancy.
- The model's weak predictive accuracy suggests that anti-displacement policies should not solely target tracts based on generic income growth figures.
- The failure of this simple model aligns with the understanding that complex systems like urban studies require algorithmic approaches, as simple parametric models imposed on complex systems result in a loss of accuracy and information.

Discussion

- **Implications for Policy:** The finding that simple income growth is not the sole, dominant driver of shifts in owner-occupancy has important implications for Dallas city planning and equity issues.
- **Policy Targeting:** Due to the lack of a strong OLS correlation, anti-displacement policies should not solely target tracts based on generic income growth figures.
- **Drivers of Transformation:** Neighborhood change in Dallas is likely driven by a combination of development types -
 - Large-scale apartment construction, which increases the renter share.
 - Targeted single-family development, which increases the owner share.
- **Predictive Factors:** A tract's characteristics from 2010 are likely more predictive of future change than the rate of income change alone.

Future Research

- **Improving Model Specification:**

- Future Ordinary Least Squares (OLS) models should incorporate initial 2010 housing value and owner share as independent variables.
- These initial conditions are considered powerful predictors of subsequent change and could help explain the variance that the income change variable failed to capture.
- This approach aligns with the Algorithmic Modeling Culture which prioritizes building complex predictive models to extract reliable information about underlying data mechanisms, recognizing that achieving high predictive accuracy generally requires complex prediction methods.

- **Refining Displacement Measurement**

- The prior analysis utilized owner share as a proxy for demographic change.
- A more robust future study must focus on the percentage point of change of specific racial or ethnic groups.
 - This refined focus would directly investigate displacement risk among minority communities, particularly addressing how urban changes disproportionately affect Latino and Black communities facing heightened risk of displacement and cultural erasure in Dallas.

- **Analyzing Spatial Heterogeneity**

- Employing Geographically Weighted Regression (GWR) is recommended.
- GWR would allow the relationship between income change and owner share change to vary spatially.
- This method identifies localized pockets where the relationship is strong, avoiding the issue of forcing a single, weak relationship across the entire Dallas County area.
- This emphasis on local variation helps visualize and highlight the heterogeneity in impacts across specific vulnerable neighborhoods, such as West Dallas, South Dallas, and Vickery Meadow

Limitations

Correlation vs Causation

- The OLS model employed identifies associations but is fundamentally unable to establish causation.
- Even if a correlation is strong, the analysis cannot definitively state whether a change in income caused a change in the owner's share, or vice versa.
- It is possible that the relationship observed between variables is actually driven by a third, unmodeled variable.
- In the social sciences, researchers often focus on identifying causal factors as their ultimate role.

Ecological Fallacy

- By performing regression and descriptive statistics at the census tract level, the analysis risks committing the ecological fallacy.
- Tract-level averages do not necessarily reflect the experience of individuals within those specific tracts.
 - For example, a rising median income average at the tract level could be the result of a small number of very wealthy individuals moving in, rather than a rise in income for the existing residents

Granularity Limits

- The analysis used Metropolitan Statistical Area (MSA)-level data from FRED/Zillow to provide a macro historical context for housing costs.
- This MSA-level data is a major limitation for localized gentrification studies because it does not capture the essential tract-to-tract variation in housing costs.
- The study used MSA-level data for the Dallas-Plano-Irving, TX MSA time-series perspective, which required filtering or disaggregation to apply to City of Dallas tracts.
- To improve future research, the prioritization of acquiring tract-level housing value data across the entire study period is recommended.

Questions?

Link to Github Repositories:

<https://github.com/Miguelgut1/US-Census---Final-Project>

[https://github.com/htoisanroi/Team-Project-Methods-of-Data-
Production-](https://github.com/htoisanroi/Team-Project-Methods-of-Data-Production-)